Programming Lab 1 Assignment

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1. Write a Python program that asks the user for their name and age. Then, print a message saying "Hello, [name]! You will be [age+10] years old in 10 years."
2. Write a program that asks the user to enter a year. The program should determine if the year is a leap year or not, and print the result. Use the following rules to determine a leap year:
   - If the year is divisible by 4, it is a leap year, except:
   - If the year is also divisible by 100, it is not a leap year, unless:
   - If the year is also divisible by 400, then it is a leap year.

   For example, 2000 and 2020 were leap years, but 1900 was not.
3. Write a program that creates a list of 5 fruits. Use a for loop to print each fruit in the list, along with its index. Then, ask the user to input a new fruit and add it to the end of the list. Finally, print the updated list.
4. Implement a simple guessing game. Generate a random number between 1 and 100, then ask the user to guess the number. Use a while loop to continue asking for guesses until the user guesses correctly. Provide "higher" or "lower" hints after each incorrect guess.
5. Create a program that does the following:
   - Ask the user how many numbers they want to enter.
   - Use a for loop to collect that many numbers from the user and store them in a list.
   - Calculate and print the sum and average of the numbers.
   - Use an if-else statement to print whether the average is above or below 50.
   - Use a while loop to keep asking the user if they want to run the program again, stopping if they enter 'no'.
6. Write a Python function to check if a given number is prime. Then, create a program that does the following:
   - Ask the user to enter a positive integer greater than 1.
   - Use a while loop to ensure the user enters a valid number.
   - Call the prime-checking function to determine if the number is prime.
   - Print whether the number is prime or not.
   - If the number is not prime, print its smallest factor other than 1.

Instructions

- Upload your assignment on Moodle before the deadline.
- Include comments in your code to explain your logic.
- **Naming format of file:** your roll number_subject code_lab number (e.g., 23M1500_IE509_lab1)
- If you use any external libraries, mention them at the beginning of your script.
- Do not copy from others; ensure your work is original.
- Explore random and maths module in python